

# UNIRATIONALITY OF THE GENERAL $(2, 3)$ HYPERSURFACE IN $\mathbf{P}^2 \times \mathbf{P}^2$ BY TANGENT RESIDUALS: THE CONSTRUCTION AND ITS VERIFICATION ARTIFACTS

ABSTRACT. Let  $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$  be a general hypersurface of bidegree  $(2, 3)$  over  $\mathbf{C}$ . Projection to the second factor makes  $X$  a standard conic bundle over  $\mathbf{P}^2$  with smooth discriminant curve of degree 9 — the smallest discriminant degree for which unirationality of conic bundles over  $\mathbf{P}^2$  was not previously known, and a case that published heuristics expected to be non-unirational. We give a complete account of an explicit construction proving that  $X$  is unirational of degree 20: the vertical surface  $S = X \cap (\mathbf{P}^2 \times L)$ , rational by Tsen’s theorem, is mapped birationally onto a *horizontal* divisor  $T \sim 10H_x + H_y$  by taking fiberwise tangent residuals in the second fibration of  $X$ , whose fibers are plane cubics. The key algebraic input is a universal resultant identity producing a covariant line through the three tangent residuals. Combined with known results, the construction yields smooth threefolds that are unirational, irrational, and not stably rational, with trivial Brauer group and trivial degree-three unramified cohomology. These notes are self-contained on the mathematics and additionally document every machine computation the proof relies on, with pointers to the certificate files in the accompanying repository.

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## 1. INTRODUCTION AND STATEMENT

A *standard conic bundle* over  $\mathbf{P}^2$  is a smooth projective threefold  $X$  with a flat morphism  $\pi: X \rightarrow \mathbf{P}^2$  whose fibers are plane conics, with relative Picard rank one. Its discriminant  $\Delta \subset \mathbf{P}^2$  is a curve; the birational geometry of  $X$  is largely governed by  $\deg \Delta$ . Rationality is

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classical:  $X$  is rational iff  $\deg \Delta \leq 4$ , or  $\deg \Delta = 5$  with even theta-characteristic (Beauville, Shokurov; see [11] for a survey), and irrational for smooth  $\Delta$  of degree  $\geq 6$  by the intermediate Jacobian/Prym criterion [2, Théorème 4.9]. *Unirationality* is far more delicate. The published record reaches discriminant degree 8: conic bundles over  $\mathbf{P}^2$  with  $\deg \Delta \leq 8$  are unirational (Kollár–Mella [8] together with [11, §14.3]; also the preprint [10], which further covers degree 9 with *singular* discriminant). For large general discriminants the expectation in the literature was negative: “*It is likely that a conic bundle with sufficiently large and general discriminant curve is not unirational*” [11, §14.3], echoed in [9].

The general hypersurface  $X$  of bidegree  $(2, 3)$  in  $\mathbf{P}^2 \times \mathbf{P}^2$  is a standard conic bundle over the second factor with *smooth* discriminant of degree 9: precisely the first case beyond the published boundary. These notes give a complete, explicit proof of:

**Theorem 1.1.** *The general hypersurface  $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$  of bidegree  $(2, 3)$  over  $\mathbf{C}$  is unirational: there is a dominant rational map  $\mathbf{P}^3 \dashrightarrow X$  of degree 20, produced by an explicit construction (Construction 3.1).*

**Corollary 1.2.** *The very general  $(2, 3)$  hypersurface  $X$  is simultaneously unirational (Theorem 1.1), irrational [2, Théorème 4.9], and not stably rational [3, Theorem 1.1]. Moreover  $\text{Br}(X) = 0$  and  $H_{\text{nr}}^3(X, \mathbf{Q}/\mathbf{Z}(2)) = 0$  [5], so the failure of stable rationality is invisible to the classical unramified invariants; unlike the Artin–Mumford threefold [1], whose non-stable-rationality is detected by  $\text{Br}$ , these examples are detected only by degeneration of the diagonal [3].*

The construction is elementary in style and could have been carried out classically; the nearest antecedent is the 1980 note of Del Centina–Verdi [6], which asserted a *special* smooth unirational  $(3, 2)$  example by an entirely different method (rationality of one double plane with special branch sextic). Their displayed data contain inconsistencies; a corrected pencil is certified in the repository (§4). The fiberwise tangent-residual mechanism used here appears to be new.

**Provenance of these notes.** The construction and proof below were produced and mechanically verified in a computational project archived at [github.com/mattrobball/unirational](https://github.com/mattrobball/unirational). The mathematical exposition is self-contained; §4 is a complete ledger of the machine artifacts the proof relies on, what each verifies, and which steps rest on standard prose mathematics instead.

## 2. SETUP

**Convention 2.1.** We work over  $\mathbf{C}$  (any algebraically closed field of characteristic 0 works verbatim). Coordinates  $x = (x_0:x_1:x_2)$  on  $\mathbf{P}_x^2$  and  $y = (U:V:W)$  on  $\mathbf{P}_y^2$ . A bidegree- $(2, 3)$  form is written

$$F(x, y) = x^\top A(y) x,$$

with  $A(y)$  a symmetric  $3 \times 3$  matrix of cubic forms in  $y$ . We write  $X = \{F = 0\}$  and let  $H_x, H_y$  denote the restrictions of the two hyperplane classes. “General” always means: in a dense Zariski-open subset of the parameter space  $\mathbf{P}^{59} = \mathbf{P}H^0(\mathcal{O}_{\mathbf{P}^2 \times \mathbf{P}^2}(2, 3))$ , intersected with the smooth locus.

The threefold  $X$  carries *two* fibrations:

- $\pi = \text{pr}_y: X \rightarrow \mathbf{P}_y^2$ , a conic bundle with discriminant  $\Delta = \{\det A(y) = 0\}$  of degree 9, smooth for general  $F$ ;

- $\rho = \text{pr}_x: X \rightarrow \mathbf{P}_x^2$ , a fibration whose fiber over  $x$  is the plane cubic  $C_x = \{F(x, -) = 0\} \subset \mathbf{P}_y^2$ , smooth for general  $x$ .

The interplay of the two is the whole story.

**Lemma 2.2.** *For smooth  $X$ :  $\text{Pic}(X) = \mathbf{Z}H_x \oplus \mathbf{Z}H_y$  and  $K_X = \mathcal{O}_X(-1, 0)$ . Every effective divisor class  $aH_x + bH_y$  has  $a, b \geq 0$ . Writing  $f_y$  (a conic fiber of  $\pi$ ) and  $c_x$  (a cubic fiber of  $\rho$ ) for the two curve classes,*

$$H_x \cdot f_y = 2, \quad H_y \cdot f_y = 0, \quad H_x \cdot c_x = 0, \quad H_y \cdot c_x = 3.$$

*Consequently a surface  $S \in |aH_x + bH_y|$  has degree  $2a$  over  $\mathbf{P}_y^2$  and degree  $3b$  over  $\mathbf{P}_x^2$ . In particular every multisection of the conic bundle has even degree, and  $\rho$  has no rational section.*

*Proof.* Grothendieck–Lefschetz for the ample divisor  $X$  in  $\mathbf{P}^2 \times \mathbf{P}^2$  ( $\dim X = 3$ ) gives  $\text{Pic}$ ; adjunction gives  $K_X$ . Effectivity: from  $0 \rightarrow \mathcal{O}(a-2, b-3) \rightarrow \mathcal{O}(a, b) \rightarrow \mathcal{O}_X(a, b) \rightarrow 0$  and  $H^1(\mathbf{P}^2 \times \mathbf{P}^2, \mathcal{O}(m, n)) = 0$  for all  $m, n$ , one gets  $h^0(\mathcal{O}_X(a, b)) = h^0(\mathcal{O}(a, b)) - h^0(\mathcal{O}(a-2, b-3))$ , nonzero only if  $a, b \geq 0$ . The intersection numbers are immediate. The parity statement is consistent with, and geometrically explains, the nontriviality of the 2-torsion Brauer class of the generic conic (restriction–corestriction).  $\square$

*Remark 2.3.* Since every surface in the threefold  $X$  is a divisor, Lemma 2.2 constrains all of them. Adjunction gives  $K_S = ((a-1)H_x + bH_y)|_S$  for  $S \in |aH_x + bH_y|$ ; for a *horizontal* class ( $a \geq 1$ ) this is effective or trivial, so *no smooth member of any horizontal linear system is rational*. Any horizontal surface with rational normalization — which Theorem 1.1 requires, by Proposition 2.4 below — is necessarily a singular member of its system. This is the precise sense in which the problem resists naive searches, and why the construction below must produce a highly singular divisor.

**Proposition 2.4** (Multisection principle). *Let  $T \subset X$  be an irreducible surface dominating  $\mathbf{P}_y^2$  with rational normalization  $\tilde{T}$ , of degree  $N$  over  $\mathbf{P}_y^2$ . Then  $X$  is unirational of degree  $N$ . Conversely, if  $X$  is unirational it contains such surfaces.*

*Proof.* Base-change:  $X \times_{\mathbf{P}_y^2} \tilde{T} \rightarrow \tilde{T}$  is a conic bundle over a rational surface carrying the tautological section  $t \mapsto (t, t)$ . A conic with a rational point over the field  $\mathbf{C}(\tilde{T})$  is isomorphic to  $\mathbf{P}^1$ , so  $X \times_{\mathbf{P}_y^2} \tilde{T}$  is birational to  $\tilde{T} \times \mathbf{P}^1$ , a rational threefold; the projection to  $X$  is dominant of degree  $N$ . For the converse, the image of a general plane under a dominant map  $\mathbf{P}^3 \dashrightarrow X$  is a horizontal surface with rational (indeed unirational, hence by Castelnuovo rational) normalization.  $\square$

Harris’s formulation of the problem — find rational surfaces in  $X$  *other than* the preimages of rational curves in  $\mathbf{P}_y^2$  [7, p. 24] — asks exactly for the horizontality in Proposition 2.4: the preimage  $\pi^{-1}(\text{curve})$  is vertical and useless. The point of the construction below is that one of these “useless” vertical surfaces can be *converted* into a horizontal one using the second fibration  $\rho$ .

### 3. THE CONSTRUCTION

**Construction 3.1.** Fix a general line  $L \subset \mathbf{P}_y^2$  and choose coordinates with  $L = \{W = 0\}$ .

- (I) **The vertical seed.** Let  $S = X \cap (\mathbf{P}_x^2 \times L) \in |\mathcal{O}_X(0, 1)|$ . Projection to  $L \simeq \mathbf{P}^1$  makes  $S$  a conic bundle over  $\mathbf{P}^1$ ; its generic fiber is a smooth conic over the  $C_1$  field  $\mathbf{C}(L) \simeq \mathbf{C}(t)$ , hence has a rational point by Tsen’s theorem, and  $S$  is a *rational*

surface (irreducible, since the generic fiber is geometrically integral). Under the other projection,  $S$  is a *trisection* of  $\rho$ : for general  $x$ , the fiber is  $C_x \cap L = \{p_1, p_2, p_3\}$ .

- (II) **Tangent residuals.** For general  $x$  the cubic  $C_x$  is smooth at each  $p_i$  (this needs only that the binary cubic  $g_x = F(x, -)|_L$  has distinct roots). Define

$$\tau: S \dashrightarrow X, \quad (x, p) \mapsto (x, \tau_x(p)), \quad T_p C_x \cdot C_x = 2p + \tau_x(p).$$

In the group law of  $C_x$  (any flex as origin) this is  $p \mapsto -2p$ .

- (III) **The horizontal image.** Let  $T = \overline{\tau(S)}$ . Theorem 3.4 below shows  $T \in |10H_x + H_y|$  and  $\tau: S \dashrightarrow T$  is birational. By Lemma 2.2,  $T$  has degree  $T \cdot f_y = 20$  over  $\mathbf{P}_y^2$ ; it is horizontal with rational normalization, so Proposition 2.4 gives a dominant degree-20 map  $\tilde{T} \times \mathbf{P}^1 \dashrightarrow X$ , proving Theorem 1.1.

Everything therefore rests on identifying the image divisor exactly. This is done by a universal polynomial identity.

**3.1. The universal resultant identity.** Write a ternary cubic in the coordinates adapted to  $L = \{W = 0\}$  as

$$(1) \quad f(U, V, W) = g(U, V) + W h(U, V) + W^2(iU + jV) + kW^3,$$

with  $g$  a binary cubic and  $h$  a binary quadratic. For  $p = [u:v:0]$  with  $g(u, v) = 0$ , the tangent line to  $\{f = 0\}$  at  $p$ , evaluated at  $[U:V:W]$ , is

$$P_{U,V,W}(u, v) = U g_u(u, v) + V g_v(u, v) + W h(u, v),$$

because  $f_U, f_V, f_W$  restrict on  $W = 0$  to  $g_u, g_v, h$ . Set

$$\mathcal{R}_f(U, V, W) = \text{Res}_{u,v}(g, P_{U,V,W}),$$

the resultant of binary forms of degrees 3 and 2; up to the standard normalization,  $\mathcal{R}_f$  is the product of the three tangent lines at the points of  $\{g = 0\}$ , a cubic in  $(U, V, W)$ .

**Theorem 3.2** (Universal identity). *Let  $\Delta(g)$  denote the classical discriminant of the binary cubic  $(18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2)$  for  $g = au^3 + bu^2v + cuv^2 + dv^3$ . Then, identically in the ten coefficients of  $f$ ,*

$$(2) \quad \mathcal{R}_f + \Delta(g) f = W^2 q_f,$$

where  $q_f$  is a linear form in  $(U, V, W)$  whose three coefficients are homogeneous of degree 5 in the coefficients of  $f$  and have greatest common divisor 1 in the universal coefficient ring.

*Proof.* Expand the left side in  $W$ . The  $W^0$  term is  $\text{Res}(g, U g_u + V g_v) + \Delta(g)g(U, V)$ . By Euler's relation  $u g_u + v g_v = 3g$ , each factor  $U g_u(p_i) + V g_v(p_i)$  of the resultant is a linear form vanishing at the root  $p_i$ , so the product is proportional to  $g(U, V)$ ; the constant is  $-\Delta(g)$  (a classical covariant identity, verified symbolically in the artifacts). The  $W^1$  coefficient of  $\mathcal{R}_f$  is a binary quadratic in  $(U, V)$ ; evaluating at a root  $p_k$  of  $g$  kills every term of the root-expansion except the one attached to  $p_k$ , and comparison at the three roots identifies it with  $-\Delta(g)h(U, V)$  (two binary quadratics agreeing at three distinct points are equal; the case  $\Delta(g) = 0$  follows by continuity of a polynomial identity). Hence the left side of (2) is divisible by  $W^2$ , and being cubic, the quotient  $q_f$  is linear. The coefficient degrees:  $\text{Res}_{3,2}$  has bidegree (2, 3) in the coefficients of its two arguments, and the coefficients of  $P$  are linear in those of  $f$ , giving degree  $2 + 3 = 5$ ; likewise  $\deg \Delta(g) \cdot f = 4 + 1$ . Primitivity ( $\gcd = 1$ ) is a machine fact (§4, artifact A1/A4).  $\square$

*Remark 3.3* (Geometric content). Restricted to the cubic  $C = \{f = 0\}$ , the product of the three tangent lines has divisor  $\sum_i (2p_i + r_i)$  where  $r_i = \tau(p_i)$ , while  $W$  has divisor  $p_1 + p_2 + p_3$ . On the cubic,  $2p_i + r_i \sim H_y|_C$  and  $\sum p_i \sim H_y|_C$ , whence

$$r_1 + r_2 + r_3 \sim 3H_y|_C - 2H_y|_C = H_y|_C.$$

A degree-3 effective divisor on a smooth plane cubic lying in  $|H_y|_C|$  is a line section: *the three tangent residuals are collinear*, and  $\{q_f = 0\}$  is the line through them. (Group-law formulation: the residual map is  $p \mapsto -2p$  and  $(-2p_1) + (-2p_2) + (-2p_3) = -2(p_1 + p_2 + p_3) = 0$ .)

**3.2. The image divisor and its class.** Now let the coefficients of  $f = F(x, -)$  vary: they are *quadratic* forms in  $x$ . Then  $\Delta(g_x)$  has bidegree  $(8, 0)$ ,  $\mathcal{R}$  has bidegree  $(10, 3)$ , and the covariant line  $q = q_{F(x, -)}(y)$  has bidegree  $(10, 1)$  — linear in  $y$ , with three coefficient forms  $q_U, q_V, q_W \in \mathbf{C}[x]_{10}$ .

**Theorem 3.4.** *For general  $F$ :*

- (1) *On  $X$  the identity (2) reads  $\mathcal{R} = W^2q$ , giving the exact divisor equality*  
 $\operatorname{div}_X(\mathcal{R}) = 2S + T_q, \quad T_q := \operatorname{div}_X(q) \in |10H_x + H_y|, \quad (10, 3) = 2(0, 1) + (10, 1).$
- (2)  *$T_q$  has no component vertical over  $\mathbf{P}_x^2$ ; its fiber over a general  $x$  is exactly the three distinct tangent residuals  $\{r_1, r_2, r_3\}$ , reduced.*
- (3)  *$T_q = T = \tau(S)$  is irreducible and  $\tau: S \dashrightarrow T$  is birational.*

*Proof.* (1)  $q|_X \not\equiv 0$  since bidegree  $(10, 1)$  admits no factor of bidegree  $(2, 3)$ , and then  $\mathcal{R}|_X = W^2q|_X \not\equiv 0$ ;  $W$  cuts  $S$  (reduced and irreducible for general  $F, L$ ).

(2) A component of  $T_q$  vertical over  $\mathbf{P}_x^2$  (i.e. of class  $(a, 0)$ ) would force  $q(x, -) \equiv 0$  along a curve of  $x$ 's, i.e. a common factor of  $q_U, q_V, q_W$ . For each potential common-factor degree  $e \geq 1$ , the locus of coefficient triples with such a factor is the image of the proper multiplication map  $\mathbf{P}(\mathbf{C}[x]_e) \times \mathbf{P}(\mathbf{C}[x]_{10-e})^3 \rightarrow \mathbf{P}(\mathbf{C}[x]_{10})^3$ -incidence, hence closed; there are finitely many  $e$ ; so  $\{F : \gcd(q_U, q_V, q_W) = 1\}$  is open. It is nonempty by an explicit witness (artifact A1/A4), hence dense. Then  $q(x, -) \equiv 0$  for only finitely many  $x$ , and no vertical divisorial component exists. For the fiber: over general  $x$ ,  $C_x \cap \{q(x, -) = 0\}$  is three points by Bézout, and by Remark 3.3 the three residuals  $r_i$  lie there; that they are pairwise distinct and off  $L$  for general  $x$  is an open condition on  $F$  satisfied at an explicit witness (see Lemma 3.5 and artifact A1/A4), so the fiber is exactly  $\{r_1, r_2, r_3\}$ , reduced.

(3)  $\tau(S) \subseteq T_q$  by (2); both have degree 3 over  $\mathbf{P}_x^2$  ( $T_q \cdot c_x = 3$  by Lemma 2.2), and by (2) the generic fiber of  $T_q$  is exactly the injective image of the generic fiber of  $S$ . Hence  $T_q$  is the closure of the image of the irreducible surface  $S$  (irreducible), and  $\tau$  is generically injective of degree 1.  $\square$

**Lemma 3.5** (Openness by properness). *Let  $\mathcal{U} \subset \mathbf{P}^{59} \times \mathbf{P}_x^2$  be the open locus of pairs  $(F, x)$  where  $g_x$  has distinct roots,  $q_{F(x, -)} \neq 0$ , the three residuals are pairwise distinct, and none lies on  $L$ . Then  $\{F : \exists x, (F, x) \in \mathcal{U}\}$  is open in  $\mathbf{P}^{59}$ : it is the complement of the image of the closed set  $(\mathbf{P}^{59} \times \mathbf{P}^2) \setminus \mathcal{U}$  under the proper projection to  $\mathbf{P}^{59}$ . A single explicit pair  $(F_0, x_0) \in \mathcal{U}$  therefore certifies the condition for general  $F$ . Concretely, all four conditions are checked at a witness by:  $\operatorname{disc}(g_{x_0}) \neq 0$ ;  $q \neq 0$ ;  $\operatorname{disc}(F(x_0, -)|_{\{q=0\}}) \neq 0$  (three distinct residuals, no root extraction needed); and  $F(x_0, -) \neq 0$  at the point  $\{q = 0\} \cap L$ .  $\square$*

*Remark 3.6* (Independent Chow-theoretic check of the class). In  $\mathbf{P}_x^2 \times L_p \times \mathbf{P}_r^2$  with hyperplane classes  $H, P, R$ , the incidence  $\{p \in C_x \cap L, r \in T_p C_x \cap C_x\}$  is a complete intersection of classes

$(2H+3P), (2H+2P+R), (2H+3R)$ ; pushing down  $L_p$  gives  $20H^2+36HR+9R^2$ . Subtracting twice the pushed diagonal (tangency multiplicity 2) leaves

$$20H^2 + 32HR + 3R^2 = (2H + 3R)(10H + R),$$

i.e. the residual cycle is a divisor of class  $(10, 1)$  on the  $(2, 3)$  hypersurface — the same answer by an independent route.

**3.3. The explicit parametrization.** For computations (and for the external verification of §4) it is convenient to have the composite map in closed staged form. With  $L = \{W = 0\}$ , parametrize  $L$  by  $y(t) = (t:1:0)$ , and suppose (after the harmless linear specialization described in §4, A4) that  $e = (1:0:0)$  lies on every conic  $F(-, y(t))$ . Writing  $A_{ij}(t)$  for the restricted coefficients of the conic:

$$(I) \quad x^*(t, m) = (A_{11} + mA_{12} + m^2A_{22}, -(A_{01} + mA_{02}), -m(A_{01} + mA_{02}))$$

is the second intersection of the chord of direction  $(0, 1, m)$  through  $e$  — a birational parametrization of  $S$  by  $(t, m)$ .

$$(II) \quad f^* = F(x^*(t, m), -), \quad (\alpha, \beta, \gamma) = \nabla_y f^*|_{y(t)}, \quad z^* = (\gamma, 0, -\alpha),$$

$$f^*(sy(t) + uz^*) = u^2(A s + B u), \quad r(t, m) = B y(t) - A z^*$$

( $z^*$  is a second point of the tangent line, since  $\alpha\gamma + \gamma(-\alpha) = 0$ ; the two leading coefficients vanish identically, expressing tangency, and  $r$  is the residual).

$$(III) \quad d = (0, 1, n), \quad Q = F(-, r), \quad x'(t, m, n) = Q(d)x^* - (Q(x^*+d) - Q(x^*) - Q(d))d.$$

Then  $\Phi(t, m, n) = (x'(t, m, n); r(t, m))$  is a dominant rational map  $\mathbf{A}^3 \dashrightarrow X$  of degree 20. Stage (III) is the chord construction on the conic fiber through the marked point  $x^*$ , and its correctness is the one-line *reflection identity* for any ternary quadratic form  $Q$  with polarization  $B_Q$ :

$$Q(Q(d)x - B_Q(x, d)d) = Q(d)^2 Q(x),$$

so  $F(x', r) = Q(d)^2 F(x^*, r) = 0$ . The degree over  $X$  equals  $\deg(T/\mathbf{P}_y^2) = T \cdot f_y = 20$ .

*Remark 3.7* (Scope). The construction uses the plane-cubic fibration  $\rho$ , which exists precisely because the second degree is 3. It therefore says nothing about arbitrary standard conic bundles with smooth degree-9 discriminant (the matrix entries need not be all cubic), nor about bidegrees  $(2, d)$ ,  $d \geq 4$ , where the fibers of  $\rho$  have genus  $> 1$  and the tangent line meets the fiber in  $d - 2 > 1$  residual points (a correspondence, not a map). Harris's question for  $d$  large [7, p. 24] remains open.

*Remark 3.8* (Byproducts in the repository). Two independent negative results certified in the same project delimit how sharp the construction is: on a general  $X$ , *every* member of  $|H_x|$  is a  $K3$  double plane (its branch sextic never acquires a point of multiplicity  $\geq 3$ ), and *no* member of  $|H_x + H_y|$  has rational normalization. So the minimal-class searches fail provably, and the first horizontal rational surface found lives in  $|10H_x + H_y|$  — a singular member, as forced by adjunction (Lemma 2.2 ff.).

#### 4. VERIFICATION ARTIFACTS

The proof above is complete as mathematics except where it invokes three kinds of finite algebraic facts: the universal identity (Theorem 3.2), the primitivity of  $q$ , and the existence of explicit witnesses (Lemma 3.5). These are machine-checked. This section is a complete



ledger. All paths are relative to the repository `github.com/mattrobball/unirational`; logs of every run are committed beside the scripts.

**A1 — the certificate of record.** `problems/B-conic-bundle-multisections/certificates/tangent_residual_local_checks.py` (with `.log`):

- verifies (2) *fully symbolically* over the universal ring  $\mathbf{Z}[a_3, \dots, k]$  in ten independent coefficient symbols (the  $W^0, W^1$  coefficients of  $\mathcal{R} + \Delta f$  vanish identically; the quotient by  $W^2$  is linear);
- verifies each coefficient of  $q_f$  is homogeneous of degree 5 and that the three have gcd 1 universally, and again after a recorded substitution by ten quadratic forms (so the general (2, 3) equation has primitive  $q$ );
- supplies the smooth-fiber witness of Lemma 3.5: an explicit  $(F_0, x_0)$  with the cubic smooth (unit Gröbner bases in three charts), three residual points computed, verified to lie on the cubic, on the tangent lines, on  $\{q = 0\}$ , off  $L$ , pairwise distinct;
- records the divisor arithmetic  $(10, 3) - 2(0, 1) = (10, 1)$ , degree 20.

The full prose proof with all genericity qualifications is `certificates/tangent_residual_theorem.md`.

**A2 — independent re-derivation (audit).** During the audit of this session, the identity was re-derived from scratch in a fresh SymPy session sharing no code with A1: the  $W^0$  divisibility, the cofactor being the *classical* discriminant (both sign conventions tested), degree-5 homogeneity, and universal  $\gcd = 1$  were reconfirmed independently.

**A3 — independent end-to-end geometric check (audit).** On an explicit cubic with rational points on  $L$  ( $g = (u - v)(u - 2v)(u + v)$ , generic  $h, i, j, k$ ), the three tangent residuals were computed *from the raw geometry* (tangent lines; factoring the double root out of the restricted cubic) and verified to be distinct, off  $L$ , and to lie on  $\{q_f = 0\}$  — confirming the geometric interpretation, not merely the algebra. A second check drew a random (2, 3) form over  $\mathbf{Q}$  and verified the witness conditions of Lemma 3.5 at  $x_0 = (1, 2, 3)$ , plus  $\gcd(q_U, q_V, q_W) = 1$  by restriction to two lines.

**A4 — external random-coefficient instance.** `problems/B-conic-bundle-multisections/external_checks` a full end-to-end falsification test on coefficients drawn from `/dev/urandom` (recorded), with four coefficients zeroed so that the section  $e = (1:0:0)$  exists over  $\mathbf{Q}$  and the composite map has exact rational coefficients (a codimension-4 linear slice needed only because Tsen’s theorem is not effective over  $\mathbf{Q}$ ; over  $\mathbf{C}$  the theorem needs no such conditioning). The harness `build_explicit_map.py` constructs the staged map of §3.3 and verifies, exactly:

| check | statement                                         | method                                 |
|-------|---------------------------------------------------|----------------------------------------|
| C1    | $F(x^*, y(t)) \equiv 0$                           | symbolic                               |
| C2    | tangency (two leading coefficients vanish)        | symbolic                               |
| C3    | $f^*(r) \equiv 0$                                 | full $100 \times 53$ grid <sup>†</sup> |
| C4    | $q_{f^*}(r) \equiv 0$                             | full $62 \times 37$ grid <sup>†</sup>  |
| C5    | reflection identity                               | generic symbols                        |
| C6    | $F(\Phi) = 0$ at sample rational points           | exact evaluation                       |
| C7    | Jacobian rank 3 (dominance)                       | exact symbolic differentiation         |
| C8    | fiber over random $y_0$ : squarefree of degree 20 | resultant                              |

<sup>†</sup>Deterministic proofs: a polynomial of bidegree at most  $(D_1, D_2)$  vanishing at a  $(D_1+1) \times (D_2+1)$  grid of distinct rational nodes is zero; the bidegree bounds (99, 52) and (61, 36) are derived in the harness.

All eight pass (`run_log.txt`); the explicit map instance is `map_instance.txt`, and `README_CHECK.md` gives CAS-agnostic instructions for third-party verification with no trust in the pipeline.

**A5 — unconditioned instance (no slice).** `external_check/unconditioned_check.py` (with `.log`) addresses the natural objection that the codimension-4 slice in A4 might matter: it runs the full mechanism on a fresh entropy draw with *no* coefficient zeroed. With no rational point available, it works in the cubic étale algebra  $K = \mathbf{Q}[\theta]/(g_0)$ ,  $g_0 = F(x_0, -)|_L$ , where the three marked points live; reduction modulo the full cubic verifies each identity at all three conjugate points simultaneously. It checks: primitivity of  $q_F$ ; the witness conditions of Lemma 3.5; tangency,  $F(x_0, r) = 0$ ,  $q(r) = 0$ , and  $r \notin L$  (via  $\text{Res}(g_0, r_W) \neq 0$ ) in  $K$ ; the stage-(III) point  $F(x', r) = 0$  in  $K$ ; and the squarefree degree-20 fiber. All pass. The slice in A4 is thereby confirmed to play no role beyond supplying  $\mathbf{Q}$ -rationality of the composite parametrization (Tsen’s theorem being non-effective over  $\mathbf{Q}$ ); by  $\text{PGL}_3 \times \text{PGL}_3$ -equivariance the sliced family  $\{X \supseteq \{e\} \times L\}$  is in no special position for anything the proof uses.

**Prose-only steps.** For completeness: the steps of the proof resting on standard mathematics rather than computation are Tsen’s theorem, Bertini/irreducibility of  $S$ , Grothendieck–Lefschetz (Lemma 2.2), the group-law remark, the properness argument of Lemma 3.5, and the Chow-theoretic cross-check. Each is classical.

**Independence and audit trail.** A1 was rerun in a clean environment during the audit (Macaulay2 1.26.06 / SymPy 1.14.0) with output matching the committed logs; A2–A3 share no code with A1; A4 takes its input from an external entropy source. The literature verification (the boundary  $\deg \Delta \leq 8$ , the status of [10] as an unpublished preprint whose degree- $\leq 8$  consequence has published backing via [8, 11], the method and status of [6], and the absence of any prior statement of Theorem 1.1) was performed against primary sources on 2026-07-22.

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